[synthetic] One-hot lookup then sequence summation produces the supplied count vector.

How does this fixed token sequence become a count vector?

Concrete token-to-vector mapping
conceptual | inference | parameters: not_applicable

Use the supplied two-token vocabulary.

tok0: red	->	e_red
tok1: blue	->	e_blue

axes: sequence axis: tok0,tok1; embedding axis: red,blue

output out: tokens / utterance@tokens

lookup

INNOVATION: Sum the supplied one-hot vectors
conceptual | inference | parameters: frozen

1. E(red)=[1,0]; E(blue)=[0,1].

2. Lookup X=[E(red), E(blue)].

3. Sum sequence axis: y=[1,1].

coordinate	red	blue
E(red)	1	0
E(blue)	0	1
y	1 + 0 = 1	0 + 1 = 1

input input: tokens / utterance@tokens
output out: vector2 / utterance@sum

sum

Representation before and after
conceptual | inference | parameters: not_applicable

Retain the utterance identity; no trained attention.

before: red, blue (ordered tokens)	->	after: y=[1,1] (two-dimensional count vector)

input input: vector2 / utterance@sum

lookup [data]: ordered token IDs; when always

sum [data]: two-dimensional count vector; when always